MANIPAL INSTITUTE OF TECHNOLOGY

ICT 4442 Deep Learning Mini Project

Customer Review Intelligence System

Comparative deep learning for sentiment classification and aspect-level evidence

FINAL REPORT

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Repository: https://github.com/pavitagrawal/CRIS_DL

Abstract

This project compares four deep learning architectures for three-class sentiment classification of Amazon customer reviews and integrates the selected model into an explainable review-analysis application. MLP, TextCNN, BiLSTM, and a compact attention-based Transformer use one category-and-class stratified 70/15/15 split, a training-only tokenizer, a 20,000-token vocabulary limit, and a 200-token sequence limit. Validation macro F1 selects the architecture before final test evaluation.

BiLSTM achieved the strongest validation macro F1 at 0.6803. On the held-out 7,500-review test split it reached 67.71% accuracy and 0.6767 macro F1. Neutral reviews remained the main error source, with 55.36% recall. A later fine-tuned BiLSTM supports short product and restaurant reviews in the Streamlit application. The application adds clause-level aspect evidence, conservative explicit cues, human-review flags, editable corrections, and downloadable results.

The aspect component is intentionally separated from the overall sentiment model. Keyword profiles and phrase cues reached 59.64% recall and 89.59% mention precision on the inspected SemEval laptop test split, while a restaurant profile reached 61.73% recall and 94.09% mention precision on SemEval restaurant Phase B. These figures describe term coverage and covered-term polarity, not end-to-end ABSA accuracy. The final system is complete for the course scope and retains these limitations in the dashboard and documentation.

1 Project objectives

The system addresses two related tasks. First, it compares complementary neural architectures under a fair experimental protocol. Second, it turns the selected model into a usable review-intelligence workflow that preserves evidence and supports human correction.

- Compare MLP, TextCNN, BiLSTM, and Transformer architectures on one saved split.
- Select models using validation macro F1 and report held-out Accuracy, macro Precision, Recall, F1, training time, prediction time, and confusion matrices.
- Expose aspect mentions and their matched clauses without presenting heuristic aspect labels as a separately trained ABSA model.
- Support single-review and CSV analysis, review triage, human labels, and downloadable annotated results.
- Make data preparation, training, analysis, and verification reproducible from scripts.

2 Background and literature

Customer reviews can contain different opinions about quality, price, design, performance, battery, service, and other attributes. Document sentiment classification summarizes the review as a whole, while aspect-based sentiment analysis identifies a target and evaluates the sentiment expressed toward it. The final project treats these as separate layers because Amazon star ratings supervise document sentiment but do not provide aspect-level labels.

Kim's TextCNN motivates local phrase filters. Hochreiter and Schmidhuber's LSTM supports sequential context, while bidirectional processing and learned attention provide a sentence representation. Vaswani et al. motivate self-attention and direct token interactions. SemEval-2014 provides public laptop and restaurant aspect annotations used here to evaluate keyword coverage and covered-term polarity.

Architecture	Primary mechanism	Role in comparison
MLP	Mean-pooled token embeddings and feed-forward layers	Fast baseline
TextCNN	Parallel one-dimensional convolutions and global max pooling	Local phrase patterns
BiLSTM	Bidirectional recurrent layers with attention pooling	Long-range sequence context
Transformer	Multi-head self-attention with a learned CLS token	Global token interaction

Table 1. Roles of the four planned architectures

3 Dataset and preprocessing

3.1 Dataset construction

The reproducible benchmark samples 50,000 reviews from Amazon Reviews 2023 categories All Beauty, Books, Electronics, and Home and Kitchen. Each category contributes 12,500 reviews. Reservoir sampling operates within the first 200,000 source records per category, filters short text, removes case-insensitive exact duplicates across categories, and balances the three rating-derived sentiment classes.

Ratings 1-2 map to negative, rating 3 maps to neutral, and ratings 4-5 map to positive. These labels are consistent and scalable, but they remain proxies for the text's sentiment. A review can contain mixed language or wording that conflicts with its star rating.

Partition	Reviews	Share	Purpose
Training	35,000	70%	Model parameter learning and tokenizer fitting
Validation	7,500	15%	Epoch selection and architecture comparison
Test	7,500	15%	Final held-out evaluation

Table 2. Fixed category-and-sentiment stratified split

3.2 Text processing

The shared pipeline lowercases text, removes HTML and URLs, filters non-alphabetic noise, optionally removes stopwords, builds a vocabulary from training text only, and converts reviews to fixed-length integer sequences. The final comparison limits the vocabulary to 20,000 tokens and pads or truncates each review to 200 tokens. Padding masks prevent padded tokens from affecting MLP pooling, BiLSTM attention pooling, and Transformer attention.

3.3 Reproducibility

The dataset builder records source URLs, scan limits, sampling seed 20261021, class quotas, deduplication, split policy, and output sizes. The model runner fixes Python, NumPy, and PyTorch seed 42. The full dataset remains excluded from version control because it is large and redistributable from the documented public source.

4 Model implementation

4.1 MLP

The baseline embeds tokens, computes a padding-aware mean representation, and applies two fully connected blocks with BatchNorm, ReLU, and dropout before the three-class output layer.

4.2 TextCNN

TextCNN applies convolutional kernels of widths 2, 3, and 4 to identify local n-gram-like patterns. ReLU activations and global max pooling produce a fixed-length feature vector for classification.

4.3 BiLSTM

The BiLSTM uses two bidirectional recurrent layers. A learned attention score combines non-padding hidden states, and the resulting context vector feeds the final classifier. Gradient clipping controls recurrent training. This architecture won the validation comparison.

4.4 Transformer

The compact Transformer uses learned token embeddings, sinusoidal positions, a learned CLS token, two encoder layers, four attention heads, and a feed-forward dimension of 256. Padding masks exclude padded positions. It trains from scratch and should not be interpreted as a pretrained BERT result.

5 Experimental protocol

All four architectures receive the same encoded training, validation, and test tensors. Each uses 128-dimensional embeddings and AdamW. MLP, TextCNN, and BiLSTM use a learning rate of 5e-4, while Transformer uses 1e-4. Training runs for at most five epochs. The script saves the epoch with the highest validation macro F1, then evaluates that state on the test split exactly once for reporting.

Control	Final setting
Vocabulary	20,000 tokens fitted on training reviews only
Sequence length	200 tokens with post padding and truncation
Batch size	128
Selection metric	Validation macro F1
Test labels	Negative, neutral, positive rating proxies
Reported efficiency	Training seconds and full-test prediction seconds

Table 3. Shared experimental controls

6 Architecture comparison

Model	Epoch	Val F1	Test accuracy	Test F1	Train	Predict
MLP	5	0.6062	61.13%	0.6115	14.3 s	0.07 s
TextCNN	5	0.6065	62.01%	0.6157	215.3 s	3.22 s
BiLSTM	5	0.6803	67.71%	0.6767	1096.5 s	14.12 s
Transformer	4	0.3942	39.57%	0.3907	1042.4 s	8.20 s

Table 4. Final four-model comparison on the shared split

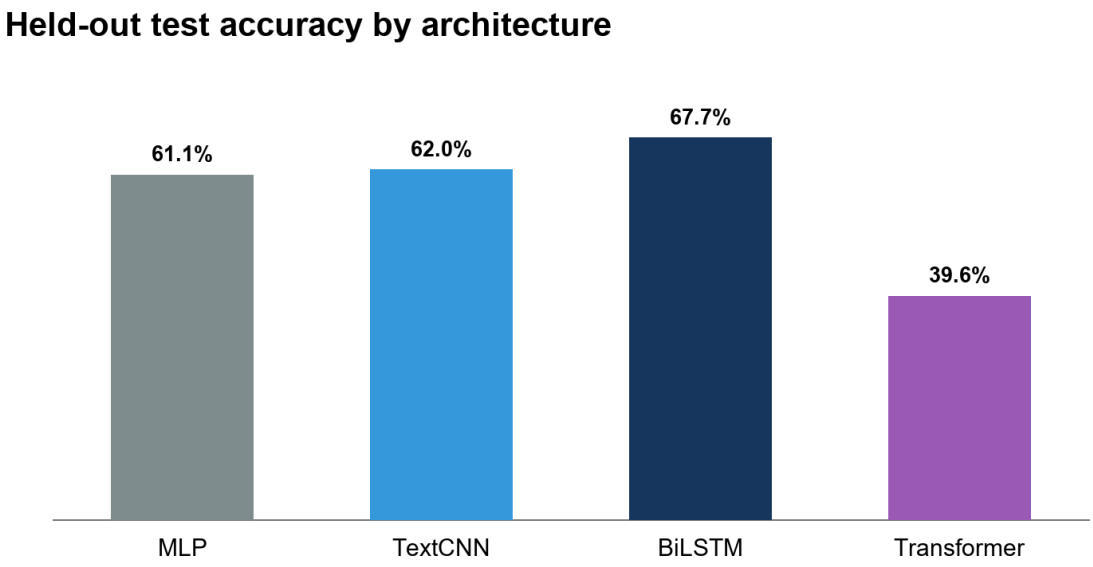


Figure 1. Held-out accuracy for the four planned architectures

BiLSTM achieved the highest validation macro F1 and therefore became the selected architecture. It also achieved the highest held-out test accuracy and macro F1. TextCNN modestly improved on MLP but required substantially more computation. The compact Transformer did not optimize successfully under the declared from-scratch configuration; its result documents this experiment rather than supporting a general claim about Transformer models.

7 Error analysis

Class	Support	Correct	Errors	Precision	Recall	F1
Negative	2500	1794	706	0.6669	0.7176	0.6913
Neutral	2500	1384	1116	0.5767	0.5536	0.5649
Positive	2500	1900	600	0.7884	0.7600	0.7739

Table 5. BiLSTM class-level test performance

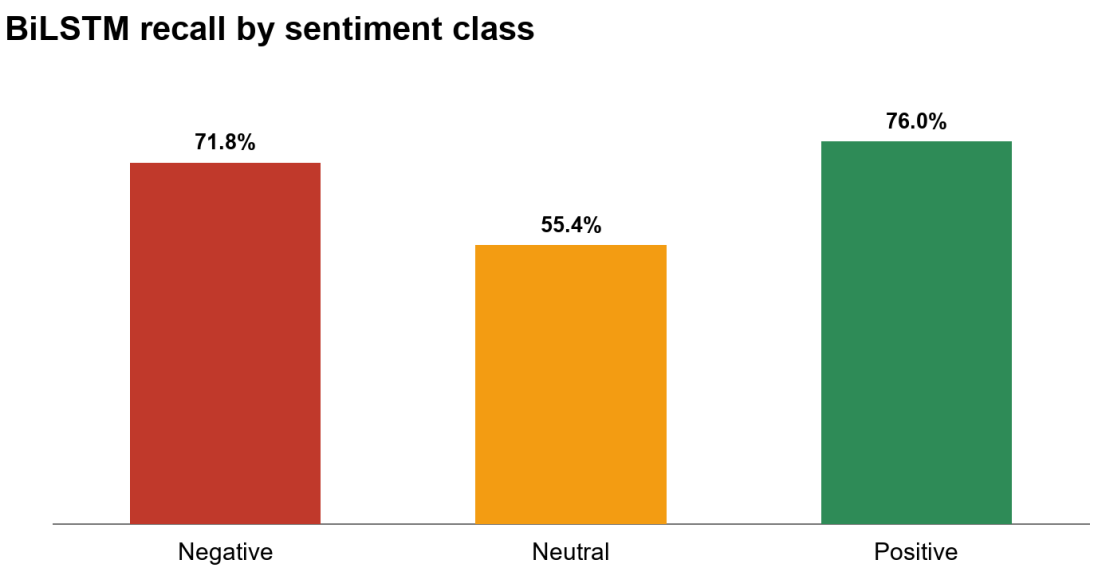


Figure 2. BiLSTM recall by sentiment class

Neutral reviews produced 1,116 of the model's 2,422 test errors. The largest individual confusion mapped 744 neutral reviews to negative. Other large confusions mapped 568 negative reviews to neutral and 448 positive reviews to neutral. This pattern explains why the application flags low-score predictions, mixed aspect evidence, and explicit complaints paired with neutral predictions for human review.

The softmax score shown by the application is uncalibrated. The review queue is a triage aid rather than a guarantee that every flagged prediction is wrong or every unflagged prediction is correct.

8 Aspect-level evidence

8.1 Method

The aspect layer segments a review into short clauses, matches category-specific keywords, and assigns each matched clause a sentiment label. Conservative phrase rules handle clear statements such as battery drains quickly or the keyboard is comfortable. Every result includes the matched clause and the method used. Aspect summaries remain separate from the overall model prediction.

8.2 Laptop evaluation

Measure	Result
Gold laptop terms	332
Detected gold terms	198
Mention precision	89.59%
Gold-term recall	59.64%
Mention F1	0.7161
Covered-term polarity accuracy	62.12%
Correct detected polarity per gold term	37.05%

Table 6. Expanded laptop aspect results on the inspected test split

8.3 Restaurant evaluation

The restaurant profile detected 700 of 1134 Phase B aspect terms, with 94.09% mention precision, 61.73% recall, and 0.7455 F1. Restaurant keywords were developed from restaurant training-term frequencies, so this comparison remains exploratory. The later Yelp fine-tuning improved held-out binary restaurant accuracy from 71.5% to 77.5%, while three-class neutral behavior remains weak because Yelp provides positive and negative labels only.

The system does not claim full end-to-end ABSA accuracy. Term-span overlap evaluates detection, category grouping remains heuristic, and polarity is scored only for covered terms with a located clause. These limits appear in the documentation and dashboard.

9 Application and human review

The Streamlit application serves the included fine-tuned PyTorch BiLSTM. Users can analyze one review or upload a UTF-8 CSV, select the review column, inspect overall sentiment and model score, view aspect evidence, compare sentiment distributions, and download annotated results. Empty rows are skipped and the upload signature prevents stale results from carrying across changed files or aspect profiles.

For uploaded datasets, the application provides a fixed random human-evaluation sample and a review queue. Users can enter three-class labels without seeing the model label in the evaluation editor, calculate accuracy and macro F1, inspect a confusion matrix, correct individual results, and export both model outputs and human corrections. No browser-session correction silently retrains the model.

Application function	Evidence and safeguard
Single review	Overall label, uncalibrated score, aspect clauses, cue disclosure
CSV analysis	Batch predictions, distributions, aspect counts, annotated download
Review queue	Low score, mixed aspects, disagreement, or explicit complaint checks
Human evaluation	Random sample, hidden model answer, accuracy, macro F1, confusion matrix
Human correction	Preserves original prediction and records final label source
Model comparison	Saved architecture and fine-tuning evidence with limitations

Table 7. Final application scope

10 Validation and limitations

The project includes unit tests for label mapping, macro metrics, aspect clauses, keyword boundaries, explicit cues, restaurant profiles, review-queue reasons, human corrections, and evaluation sampling. Dependency-light validation confirms the four expected architectures, validation-selected winner, runtime checkpoint, tokenizer, required documentation, and JSON integrity.

Limitation	Impact	Control in the final system
Rating-derived labels	Text sentiment can disagree with stars	Describe results as proxy-label agreement
Neutral weakness	Mixed and factual reviews are often misclassified	Macro metrics, class analysis, and review queue
Aspect heuristics	Unseen expressions and categories remain uncovered	Evidence clauses, profile selector, abstention, and measured coverage
Uncalibrated scores	Softmax values are not correctness probabilities	Dashboard disclosure and no probability claim
Dataset not stored in Git	Training requires regeneration or separate bundle	Deterministic builder and provenance
From-scratch Transformer	Weak result does not represent pretrained models	Explicit interpretation limit

Table 8. Final limitations and controls

11 Reproduction guide

On Windows, run scripts/bootstrap.ps1 to create a local virtual environment and install dependencies. Run scripts/reproduce_phase3.ps1 to download and sample the four categories, recreate the fixed split, train all four models, generate the comparison, and rebuild the class-level analysis. Run

scripts/check.ps1 for bundle validation and the test suite. To demonstrate the included checkpoint without retraining, run python -m streamlit run app.py.

The repository keeps large downloaded datasets and non-runtime experiment checkpoints out of ordinary commits. The included runtime checkpoint and tokenizer remain eligible for version control so the final application can run from a fresh checkout.

12 Team contributions

Member	Primary project contribution
Pavit Agrawal	Project setup, preprocessing, dataset pipeline, MLP, training and evaluation
Priyanshu Sharma	TextCNN architecture, training, and experiment notebook
Shivam Lahoty	BiLSTM, Transformer, experiment notebooks, and comparative analysis
Aryan Sorout	Application integration, evaluation, visualization, and reporting support

Table 9. Contribution summary aligned with repository evidence

13 Conclusion

The project completes the Phase 3 roadmap. It implements and compares all four planned architectures under one reproducible protocol, selects BiLSTM by validation macro F1, documents class-level errors, evaluates aspect evidence on public annotations, and integrates the selected model into a usable review-analysis application. The strongest result is a transparent, reproducible system rather than an unsupported universal-accuracy claim. Neutral sentiment, proxy labels, and heuristic aspect coverage remain the main technical limits and are disclosed throughout the final deliverables.

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